Band-1 Hadamard Next Constructive Gap after H_2036 — Method Triage + Miyamoto Refresh

Next standard gap 2052; H_528 Kronecker + H_2116 Miyamoto/GF(23^2) PASS; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / literature ledger increment (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-14_Band1_Hadamard_508_to_2036_Miyamoto`; `2026-09-14_Band1_Hadamard_2036_method_t`
Lab. `/workspace/experiments/band1/hadamard_next_gap_after_2036_triage/`.
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo refreshes the constructive-gap list after on-box H_{2036}, triages the first 24 remaining standard-closure misses, and deep-dives the first Miyamoto-eligible remaining order (**2116**). It claims **finite constructions of orders 528 and 2116 only** (verified $\max|HH^\top - nI| = 0$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. `clay_closed = false`.

Abstract

After Band-1 constructed H_{508} (Goethals–Seidel + skew SDS 127) and H_{2036} (Miyamoto + Paley conference 510), the Sylvester∪Paley∪Kronecker closure of $\{\text{all } \leq 2000\} \cup \text{on-box}\{508, 2004, 2036\}$ has **next constructive gap 2052**. Among the first 24 remaining misses, Miyamoto-eligible orders are **2116, 2164, 2228, 2276** (Yamada-A also flags 2116 and 2276). Deep-dive on **2116** ($q = 529 = 23^2 \equiv 1 \pmod 4$): H_{528} constructed on box as Kronecker of Paley-I $H_{12} \otimes H_{44}$, and H_{2116} constructed via Miyamoto with a Paley conference matrix of order 530 over GF(23^2). Both verify $\max |HH^\top - nI| = 0$. Unit tests **16/16 PASS**. The **full Hadamard conjecture remains OPEN**. Standard gap **2052** remains open under SPK closure.

Keywords. Hadamard matrix; order 2116; Miyamoto; GF(23^2); constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. Standard-closure next gap after filling 2036 is **2052**. 2. H_{528} constructed and verified (Kronecker Paley-I). 3. H_{2116} constructed and verified (Miyamoto). **Not** the Hadamard conjecture.

2. Constructive closure refresh

Seeds: all admissible orders ≤ 2000 (Aug 2026 literature fill) ∪ on-box $\{508, 2004, 2036\}$ ∪ Sylvester powers of 2 ∪ Paley I/II, closed under Kronecker (limit 4000).

Quantity	Value
Known count	876
2004 / 508 / 2036 in closure	True / True / True
2116 in SPK closure	False (Miyamoto bonus, not SPK)
Next standard gap	**2052**
Missing first 24	[2052, 2060, 2076, 2092, 2108, 2116, 2148, 2156, 2164, 2172, 2212, 2228, 2276, 2284, 2292, 2300, 2316, 2324, 2332, 2356, 2380, 2396, 2420, 2444]

3. Method triage (first 24 misses)

n	Paley I	Paley II	Kronecker	Miyamoto	Yamada-A	$q = n/4$
2052	—	—	—	—	—	513

n	Paley I	Paley II	Kronecker	Miyamoto	Yamada-A	$q = n/4$
2060	—	—	—	—	—	515
2076	—	—	—	—	—	519
2092	—	—	—	—	—	523
2108	—	—	—	—	—	527
2116	—	—	—	Y	Y	529
2148	—	—	—	—	—	537
2156	—	—	—	—	—	539
2164	—	—	—	Y	—	541
2172	—	—	—	—	—	543
2212	—	—	—	—	—	553
2228	—	—	—	Y	—	557
2276	—	—	—	Y	Y	569
2284	—	—	—	—	—	571
2292	—	—	—	—	—	573
2300	—	—	—	—	—	575
2316	—	—	—	—	—	579
2324	—	—	—	—	—	581
2332	—	—	—	—	—	583
2356	—	—	—	—	—	589
2380	—	—	—	—	—	595
2396	—	—	—	—	—	599
2420	—	—	—	—	—	605
2444	—	—	—	—	—	611

Miyamoto-eligible: [2116, 2164, 2228, 2276].
Yamada-A eligible: [2116, 2276].

4. Deep-dive: order 2116

Arithmetic. $2116 = 4 \cdot 529$, $529 = 23^2$, $529 \equiv 1 \pmod 4$, $529 \equiv 1 \pmod 8$ (Yamada-A also eligible). Needs H_{528}.

H_{528} **path.** Paley I / II fail for 528. Kronecker eligible: $528 = 12 \times 44$. Both factors are Paley-I ($q = 11$ and $q = 43$). On-box: $H_{528} = H_{12} \otimes H_{44}$. Verify: $\max |HH^\top - 528I| = 0$ (**PASS**).

Conference. Symmetric Paley conference of order 530 over $\mathrm{GF}(23^2) = \mathbb{Z}/23[x]/(x^2+1)$ ($23 \equiv 3 \pmod 4$). Verify $CC^\top = 529I$, zero diagonal (**PASS**).

Miyamoto. $H_{2116} = \mathrm{Miyamoto}(H_{528}, C_{530})$. Verify: $\max |HH^\top - 2116I| = 0$ (**PASS**). Wall (deep-dive portion) $\approx 5.2455s$.

Object	Method	$\max \lvert HH^\top - nI \rvert$	Status
H_{528}	Kronecker Paley-I 12×44	0	**PASS**
C_{530}	Paley / GF(23^2)	0 (conf)	**PASS**
H_{2116}	Miyamoto	0	**PASS**

5. Unit tests

Test	Result
`H20_paley_I_control`	PASS
`H4_sylvester`	PASS
`H12_paley_I`	PASS
`miyamoto_H20_from_H4`	PASS
`gf9_chi_balanced`	PASS
`conference_10_gf9`	PASS
`kronecker_H12xH4`	PASS
`2036_in_onbox_seeds`	PASS
`508_in_onbox_seeds`	PASS
`q529_is_23_squared`	PASS
`q529_equiv_1_mod_4`	PASS
`q529_equiv_1_mod_8`	PASS
`528_kronecker_12_44`	PASS
`miyamoto_pred_n2116`	PASS
`yamada_a_pred_n2116`	PASS
`reject_all_ones_4`	PASS

All 16/16 unit tests PASS. End-to-end H_{528} and H_{2116} verifications also PASS.

6. Non-claims

- Full Hadamard conjecture: **OPEN** (Band-3).
- No Clay Millennium problem closed; `clay_closed = false`.
- GATE-04 collection: **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Standard gap **2052** is **not** claimed constructed (not Miyamoto-eligible: $q = 513$ not a prime power).
- Orders 2164 / 2228 / 2276 remain **existence-reduction candidates** only in this memo (not constructed here).

7. Reproducibility

```
cd /workspace/experiments/band1/hadamard_next_gap_after_2036_triage
python3 verify_next_gap_triage.py
```

Emits `results.json`, `literature_ledger.json`, `SUMMARY.txt`, and on success `H528_pm1.npz`, `H2116_pm1.npz`. SageMath is absent; pure NumPy.

8. Conclusion

Outcome. PASS — next standard gap **2052**; method triage refreshed; H_{528} and H_{2116} constructed and verified on box. Full Hadamard conjecture remains **OPEN**.

Next optional. Triage/construct remaining Miyamoto-eligible gaps (**2164** needs H_{540}; **2228** needs H_{556}; **2276** needs H_{568}), or method-attack standard gap **2052**, or Track B privacy / Object 1 meter await.

Shelf. `2026-09-14_Band1_Hadamard_next_gap_after_2036_triage.md` / `.pdf` under `/workspace/DOWNLOAD_PDFs/`.

Wall seconds (full probe): 5.4939.